MIL-J-26013B(USAF)
17 MARCH 1987
SUPERSEDING
MIL-J-26013A(USAF)
3 June 1957

MILITARY SPECIFICATION

JACK, AIRCRAFT, HYDRAULIC, TYPE A-5

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of aircraft axle jack, 5-ton capacity, designated type A-5.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-636	Boxes, Shipping, Fiberboard.
PPP-T-60	Tape, Packaging, Waterproof.

MILITARY

MIL-P-116	Preservation, Method of.
MIL-L-10547	Liner, Case, and Sheet, Overwrap, Water Vaporproof or Waterproof, Flexible.
MIL-H-81019	Hydraulic fluid, Petroleum Base, Ultra-low Temperature.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, TX 78241 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.
MIL-STD-808	Finish, Materials and Processes for Corrosion Prevention and Control in Support Equipment.
MIL-STD-810	Environmental Test Methods and Engineering Guidelines.
MIL-STD-831	Test Reports, Preparation of.
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing with Appropriate Test Methods.
MIL-STD-2073/1A	DOD Material Procedures for Development and Application of Packaging Requirements.

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

AIR FORCE

53D22020	Jack-Hand, Hydraulic Axle, 5-Ton Capacity, Type A-5, Assembly of.
7545352	Requirements for finishes, Protectives and Codes for Ground Support and Ground Support Equipment.

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIAL (ASTM)

ASTM D-3951	Packaging, Commercial.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets of MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in a contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 General. The requirements specified on Drawing 53D22020 are applicable as requirements of this specification. When the requirements of the drawing and this specification conflict, the requirements of this specification shall govern.

3.3 Performance.

3.3.1 Static proof. The jack shall be capable of raising a 7.5-ton load vertically from the lowest position to the fully extended position and of withstanding a horizontal load of 0.75 ton while supporting a 5-ton vertical load in the fully extended position.

3.3.2 Low temperature. The jack shall be capable of satisfactory operation at -54" C.

3.3.3 Life. The jack shall be capable of satisfactory operation after 200 cycles of raising a 5-ton load vertically, and 50 cycles of raising a 5-ton load vertically while being subjected to a horizontal load of 0.75 tons.

3.3.4 Leakage. The cylinder and ram assembly shall not leak when subjected to a hydraulic pressure of 7,500 psi.

3.3.5 By-pass valve. The by-pass valve shall be adjusted to the value specified on the drawing.

3.4 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.5 Treatment and painting. Treatment and painting shall be in accordance with MIL-STD-808 in conjunction with Drawing 7545352.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of tests. The inspection and testing of the jack shall be classified as follows:

a. Acceptance tests (see 4.2.1).

b. First article testing (see 4.4).

4.2.1 Acceptance tests. The acceptance tests shall consist of:

a. Individual tests (see 4.2.1.1).

b. Sampling test (see 4.2.1.2).

4.2.1.1 Individual tests. The jack shall be subjected to the following tests: (see 4.3).

a. Examination of product (see 4.3.1).

b. Leakage test (see 4.3.2).

c. By-pass valve test (see 4.3.3).

4.2.1.2 Sampling test. Five jacks which have been manufactured under essentially the same conditions and submitted for inspection at substantially the same time, shall be selected at random from each lot of 50 or fraction thereof on the order and subjected to the following test (see 4.3).

a. Static proof test (see 4.3.4).

4.2.2 Rejection and retest. When one or more jacks from a lot fail to meet the specification, acceptance of all jacks in the lot will be withheld until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.2.2.1 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of the entire lot shall not be made until it is determined that the lot meets all the requirements of the specification.

4.2.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

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4.3 Test methods.

4.3.1 Examination of product. The jack shall be inspected to determine compliance with the requirements specified herein with respect to workmanship, marking, and compliance with the drawing.

4.3.2 Leakage test. The cylinder and ram assembly of the assembled jack shall be tested by applying a minimum hydraulic pressure of 7,500 psi. This test shall be performed with an appropriate load on the ram assembly to preclude damage to ram stops. Any leaking of hydraulic fluid shall be cause for rejection.

4.3.3 By-pass valve test. The pump shall be tested by sealing the pump outlet and operating the pump until the by-pass valve opens. The by-pass valve shall be adjusted to the value specified on the drawing.

4.3.4 Static proof test. After the by-pass valve has been adjusted, the jack shall be tested by raising a 7.5-ton load vertically from the lowest position to the fully extended position. At the fully extended position, under a 5-ton load, a horizontal load of 0.750 ton shall be applied to the top of the extended screw extension. Failure of any part of the jack shall be cause for rejection.

4.3.5 Low temperature test. The jack assembly shall be placed in a test chamber and cooled to -54°C for a period of 72 hours. At the end of this period and at -54°C, the jack ram shall be raised to its maximum extension and pumped until the by-pass relief valve operates. The lowering valve shall be opened and the ram shall retract with not more than 50 pounds applied to the ram. For this test, the jack shall be serviced with hydraulic fluid conforming to MIL-H-81019. The time and number of full strokes necessary to fully extend the ram shall be recorded at both +21°C and -54°C. The time and strokes required to fully extend the ram at -54°C shall not be more than double the time and strokes required at +21°C. This test shall be conducted in accordance with MIL-STD-810, method 502.1.

4.3.6 Life test. The life test shall consist of 200 cycles of raising a 5-ton load vertically and 50 cycles of raising a 5-ton load vertically with a horizontal load of 0.750 ton applied to the top of the extended screw extension. One cycle shall consist of raising the ram from collapsed height to the fully extended position and return to the collapsed position. The horizontal load shall be graduated from 0 in the collapsed position to 0.750 ton in the fully extended position. Failure of any part of the jack during this test shall be cause for rejection.

4.4 First article testing.

4.4.1 First article testing. First article testing is required to determine conformance of the jack to the drawing with respect to materials, workmanship, and manufacturing techniques to insure that the functional capabilities of the jack will meet the design objective upon which the drawing is based.

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4.4.2 First article test sample. The first article test sample shall consist of one jack representative of the production jack. It shall be tested at a laboratory designated by the procuring activity or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

4.4.3 First article test report. If the testing is accomplished by the contractor, he shall prepare a first srticle test report according to MIL-STD-831 and furnish three copies of the report to the procuring activity.

4.4.4 First article tests. First article tests shall consist of all tests described under 4.3.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, C or Commercial as specified (see 6.2).

5.1.1 Level A. The jack shall be preserved in accordance with MIL-P-116, method I.

5.1.2 Level C. The jack shall be preserved according to MIL-STD-2073 for this level.

5.1.3 Commercial. The jack shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, C or commercial as specified (see 6.2).

5.2.1 Level A. The jack preserved and packaged to meet 5.1.1 shall be packed in exterior shipping containers conforming to MIL-STD-2073. As far as practical, exterior containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required, and contain identical quantities. The gross weight of each pack shall be limited to approximately 200 pounds. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. Containers shall be provided with a caseliner conforming to MIL-L-10547, and shall be sealed in accordance with the appendix thereto. The case liner will not be required when the unit, intermediate, or exterior container conforms to PPP-B-636 and is sealed at all joints and seams, including manufacturer's joint, with tape conforming to PPP-T-60.

5.2.2 Level B. The jack preserved and packaged to meet 5.1.1 shall be packed in domestic-type exterior containers conforming to MIL-STD-2073. Exterior containers shall be of minimum cube and tare consistent with the protection required. As far as practical, exterior containers shall be of uniform size and shape and contain identical quantities. The gross weight of each pack shall be limited to approximately 500 pounds. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto.

5.2.3 Level C. Packing for this jack shall conform to the MIL-STD-2073 requirements for this level.

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5.2.4 Commercial. The jack, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Physical protection. Cushioning, blocking, bracing and bolting as required shall be in accordance with MIL-STD-1186, except that for domestic shipments, waterproofing requirements for cushioning materials and containers shall be waived. The drop tests of MIL-STD-1186 shall be waived when preservation, packaging, and packing of the item is for immediate use or when the drop tests of MIL-P-116 are applicable.

5.4 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129, or ASTM D 3951 as applicable.

6. NOTES

6.1 Intended use. The type A-5 jack covered by this specification is intended for use as an aircraft axle jack.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Stock number.

c. Applicable levels of preservation and packaging and packing.

6.3 First article. When a first article inspection is required, the item should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Subject term (key word) listing.

Aircraft jack, hydraulic, type A-5

Hydraulic jack, aircraft, type A-5

Jack, aircraft, hydraulic, type A-5

Type A-5, jack, aircraft, hydraulic

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6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force - 99

Preparing activity:
Air Force - 82

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-J-26013 B	Jack, Aircraft, Hydraulic, Type A-5

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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